

Predictive Maintenance using Hybrid Model

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Abstract— Predictive maintenance enables early detection of equipment failures, reducing downtime, maintenance costs, and improving machine lifespan. This project presents a machine learning-based predictive maintenance system using industrial sensor data with features such as temperature, speed, torque, and tool wear. To address class imbalance (only 3.4% failure cases), SMOTE is applied to create a balanced dataset, along with feature normalization using StandardScaler. Multiple classifiers are evaluated, and a hybrid ensemble model combining the best-performing algorithms is developed. The final model is deployed as a Flask web application for real-time failure prediction. Results demonstrate that data preprocessing, class balancing, and ensemble learning significantly enhance prediction accuracy for rare failure events.

Keywords— Predictive Maintenance, Machine Learning, SMOTE, Random Forest, Hybrid Model, Class Imbalance, Industrial IoT, Flask Deployment

I. INTRODUCTION

Modern industrial manufacturing environments rely heavily on continuous and reliable machine operation. Unexpected equipment failure can result in costly downtime, safety risks, and reduced production efficiency. Traditional maintenance approaches, such as reactive maintenance (repair after failure) and preventive maintenance (scheduled servicing), are often inefficient—either occurring too late or requiring unnecessary resource expenditure.

Predictive Maintenance (PdM) overcomes these limitations by using data-driven techniques to forecast equipment failures before they occur. With the advancement of Industry 4.0, industrial machines are equipped with sensors that continuously generate operational data. This data serves as input for machine

learning algorithms to identify patterns and provide early failure predictions, enabling timely and informed maintenance decisions.

This project focuses on developing a complete predictive maintenance system using machine sensor data. A key challenge addressed is the severe class imbalance in real-world datasets, where failure cases are very rare. To tackle this, a structured preprocessing pipeline is implemented, including Exploratory Data Analysis (EDA), SMOTE-based class balancing, and feature normalization. Multiple machine learning models are trained and evaluated, and the best-performing ones are combined into a hybrid ensemble model to improve prediction accuracy.

The final system is deployed as a web-based application using Flask, allowing real-time prediction based on user input. This makes the solution practical and accessible for industrial use. Overall, the project demonstrates how machine learning, combined with proper data preprocessing and ensemble techniques, can significantly enhance the effectiveness of predictive maintenance systems.

II. LITERATURE SURVEY

Guidotti et al.[1] present a systematic literature review of supervised machine learning techniques for predictive maintenance in Industry 4.0, analysing 216 research papers. The study highlights the widespread use of models such as neural networks, random forests, and SVM for failure prediction using sensor data. It also identifies key challenges like data scarcity, class imbalance, and limited public datasets. Additionally, the

paper emphasizes the growing importance of hybrid models and explainable AI in improving real-world deployment of predictive maintenance systems.

This paper[2] presents a comprehensive systematic review of predictive maintenance techniques using machine learning and deep learning in industrial environments. It highlights the role of IoT-based sensor data and advanced models such as CNN, LSTM, and hybrid approaches for fault detection and remaining useful life prediction. The study identifies key challenges including data scarcity, noise in sensor data, and lack of model interpretability. It also emphasizes the importance of explainable AI and hybrid models for improving reliability in real-world applications. The findings support the use of data-driven approaches for efficient predictive maintenance systems.

This paper[3] proposes an IoT-integrated machine health monitoring system using a mobile application backend. The authors compare Decision Tree, XGBoost, SVM, and KNN classifiers on sensor data from industrial machines. XGBoost achieved the highest classification accuracy and was selected for the deployed application. This paper is directly relevant to the current project, as it validates the use of classification ML models for real-time machine health prediction using sensor features similar to those in our dataset.

This paper[4] evaluates AdaBoost, Random Forest, Gradient Boosting, and SVM for maintenance prediction, achieving 0.95 accuracy with Random Forest. The Crayfish optimization algorithm is applied to improve accuracy to 0.97. This paper is highly relevant as it directly benchmarks the same class of algorithms used in our project on a similar predictive maintenance problem domain.

This theoretical paper[5] reviews the mathematical foundations of Random Forest, including resampling mechanisms, variable importance measures, and parameter tuning. It provides theoretical grounding for understanding why ensemble tree methods excel in high-dimensional classification tasks, supporting the use of Random Forest in our hybrid model framework.

This paper [6] provides a comprehensive overview of the Random Forest algorithm, its theoretical basis, and practical applications in classification and regression. The authors demonstrate RF on a credit card default classification problem. This paper is relevant as it provides foundational understanding of one of the core algorithms evaluated in our project

This paper [7] presents an application of machine learning in predictive maintenance within Industry 4.0 environments using IoT-based sensor data. The study demonstrates how models such as gradient boosting and neural networks can detect abnormal machine behavior and predict equipment faults. It also highlights the integration of machine learning systems with industrial control systems and CMMS using REST APIs for real-time maintenance actions. The results show that machine learning improves fault detection compared to traditional rule-based methods. This work supports the use of data-driven approaches for practical predictive maintenance implementation.

The author [8] present a systematic literature review of machine learning techniques applied to predictive maintenance in

industrial systems. The study analyses various approaches for failure classification, fault detection, and remaining useful life prediction using real-world sensor data. It highlights the effectiveness of traditional machine learning models in structured industrial environments and the increasing adoption of data-driven maintenance strategies. The paper also discusses different maintenance approaches, including reactive, preventive, and predictive maintenance. The findings support the use of machine learning for improving reliability and reducing downtime in industrial operations.

This paper[9] focuses on addressing the class imbalance problem in machine learning through resampling techniques such as oversampling and SMOTE. It highlights how imbalanced datasets can bias models toward the majority class, reducing prediction performance for minority events. The study reviews various resampling strategies and their impact on improving classification accuracy. It also discusses challenges such as data complexity and noise in imbalanced domains. The findings support the use of SMOTE-based techniques for enhancing model performance in rare event prediction tasks.

This paper focuses [10] on addressing the class imbalance problem in machine learning using synthetic oversampling techniques such as SMOTE. It explains how imbalanced datasets can lead to biased model performance and poor prediction of minority classes. The study proposes generating synthetic samples to improve classification accuracy and model generalization. It also evaluates the effectiveness of oversampling techniques across different datasets. The findings support the use of SMOTE for improving prediction performance in rare event classification problems.

This paper [11] proposes a predictive maintenance model using the K-Nearest Neighbors (KNN) algorithm for air navigation equipment. The study includes data preprocessing, normalization, and evaluation using metrics such as accuracy, precision, recall, and F1-score. The results show that KNN achieves high accuracy and outperforms models like Random Forest and Logistic Regression. The use of oversampling techniques further improves performance in handling imbalanced data.

This paper[12] propose a hybrid predictive maintenance framework that combines physics-based mathematical modeling with an AI-based BiLSTM network for rotating machinery systems. The study integrates sensor data such as vibration and temperature to enable accurate fault detection and remaining useful life prediction. The hybrid approach significantly outperforms standalone models in terms of accuracy and reliability. Experimental results demonstrate improved early fault detection and effective real-world industrial deployment.

A. Gap Analysis

Based on the literature review, several research gaps and opportunities are identified that the current project aims to address:

- **Class Imbalance Handling:** Most surveyed papers do not explicitly address the severe class imbalance inherent in real-world maintenance datasets. The

current project explicitly applies SMOTE to create a balanced training distribution, improving recall for the minority failure class.

- **Hybrid Ensemble Deployment:** While existing works evaluate individual classifiers, few propose and deploy a hybrid model that combines the strengths of the two best individual classifiers. Our approach fills this gap.
- **End-to-End Web Deployment:** Most literature presents only offline model evaluation. This project extends to a functional Flask web application providing real-time predictions, bridging the gap between academic model development and practical industrial deployment.
- **Standardized Preprocessing Pipeline:** The project establishes a reproducible, step-by-step preprocessing pipeline (EDA → SMOTE → Normalization) that is not consistently documented in prior works.

III. METHODOLOGY

A. Problem Statement

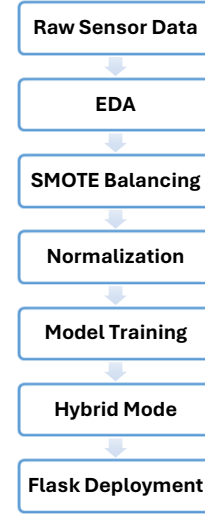
Industrial machines are subject to unexpected failures that cause significant operational disruptions and financial losses. Traditional maintenance strategies fail to leverage the rich operational sensor data continuously generated by modern machinery. The problem addressed in this project is: given a multi-feature sensor dataset of machine operating conditions, develop a machine learning model capable of accurately predicting whether a machine will fail, even in the presence of severe class imbalance where failure events constitute only 3.4% of total records.

B. Objectives

1. To perform a thorough Exploratory Data Analysis (EDA) on the Predictive Maintenance dataset to identify data distributions, correlations, outliers, and class imbalance.
2. To apply SMOTE to balance the training dataset and address the minority class underrepresentation problem.
3. To normalize all numerical features using StandardScaler to ensure scale-invariant model training.
4. To train and evaluate six machine learning classifiers: Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and Gradient Boosting.
5. To develop a hybrid ensemble model by combining the two best-performing individual classifiers.
6. To deploy the hybrid model as a Flask web application for real-time predictive maintenance inference.

C. System Architecture / Workflow

The overall system follows a structured pipeline as illustrated below:



Phase 1 covers data ingestion and preprocessing: EDA is performed on the raw imbalanced dataset, followed by SMOTE-based balancing and StandardScaler normalization to produce a clean, balanced, normalized dataset ready for machine learning. Phase 2 covers model training: six classifiers are independently trained and evaluated. The two best-performing models are combined into a hybrid ensemble (Voting or Stacking) and the final model is exported as a serialized pickle file for deployment.

D. Dataset

The project uses the AI4I 2020 Predictive Maintenance Dataset, a publicly available synthetic dataset that reflects real manufacturing sensor readings. The dataset is structured as follows:

Table 1: Dataset Description

Feature	Description	Type
Type	Machine type: L (Low), M (Medium), H (High)	Categorical
Air Temperature [K]	Ambient air temperature in Kelvin	Continuous
Process Temperature [K]	Internal process temperature in Kelvin	Continuous
Rotational Speed [rpm]	Motor rotational speed in revolutions per minute	Continuous

Torque [Nm]	Torque applied by the motor in Newton-metres	Continuous
Tool Wear [min]	Accumulated tool usage time in minutes	Continuous
Target	Binary failure label: 0 = No Failure, 1 = Failure	Binary

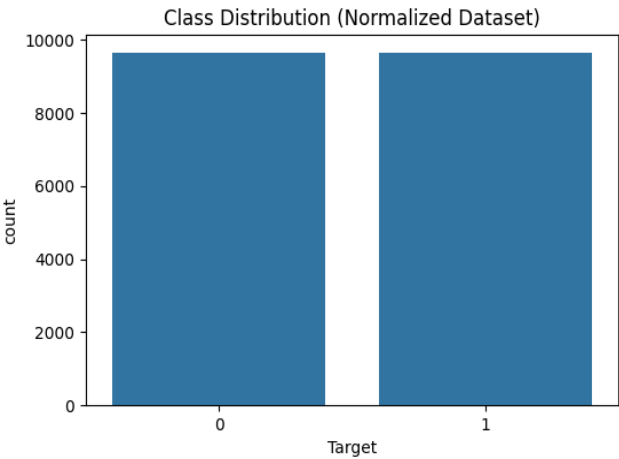


Fig. 1: Class Distribution before SMOTE (9661 No Failure vs 339 Failure)

This bar chart represents the class distribution of the dataset after normalization and preprocessing. It shows that both classes—0 (no failure) and 1 (failure)—have nearly equal counts, around 9600 instances each. This indicates that the dataset has been effectively balanced, likely using techniques such as SMOTE. A balanced dataset helps in preventing model bias toward the majority class and improves the prediction of rare failure events. Overall, this ensures better model performance and reliability in classification tasks.

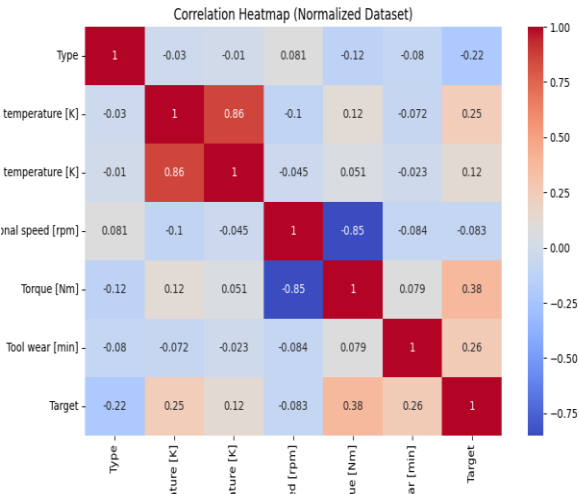


Fig. 2: Feature Correlation Heatmap

This correlation heatmap shows the relationships between different features in the normalized dataset. Strong positive correlation is observed between **air temperature and process temperature** (~ 0.86), indicating they are closely related. A strong negative correlation exists between **rotational speed and torque** (~ -0.85), meaning as one increases, the other decreases. The target variable (failure) shows moderate correlation with **torque** (~ 0.38) and **tool wear** (~ 0.26), suggesting these features significantly influence failure prediction. Most other features have weak correlations, indicating minimal redundancy. Overall, the heatmap helps in understanding feature importance and relationships for model building.

E. Data Preprocessing

1) Exploratory Data Analysis

EDA was conducted on the raw dataset to understand its structure and properties. The analysis confirmed the absence of missing values across all 10,000 records. Feature distributions were examined using distribution plots and box plots. Key findings include: (i) Air and Process temperatures exhibit near-normal distributions, (ii) Rotational speed shows a right-skewed distribution, (iii) Torque shows a normal distribution centered around 40 Nm, and (iv) Tool wear is uniformly distributed up to approximately 250 minutes. The correlation heatmap revealed moderate positive correlation between Air Temperature and Process Temperature, consistent with physical expectations.

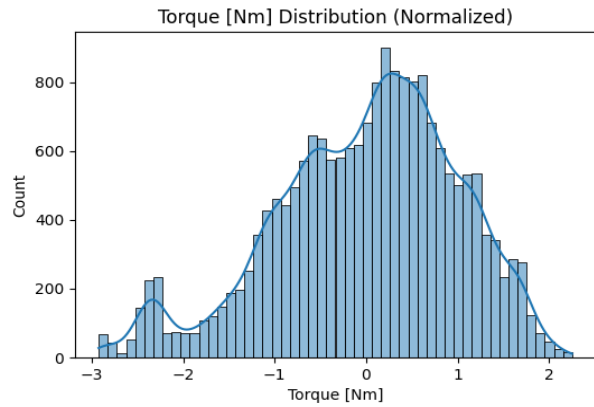


Fig. 3: Torque Distribution (Normalized Dataset)

2) Class Balancing using SMOTE

During EDA, the dataset was found to be severely imbalanced: 9,661 records belong to the No Failure class (96.6%) and only 339 records represent actual Failure events (3.4%), producing a 28:1 imbalance ratio. This is one of the most dangerous problems in real-world ML classification tasks.

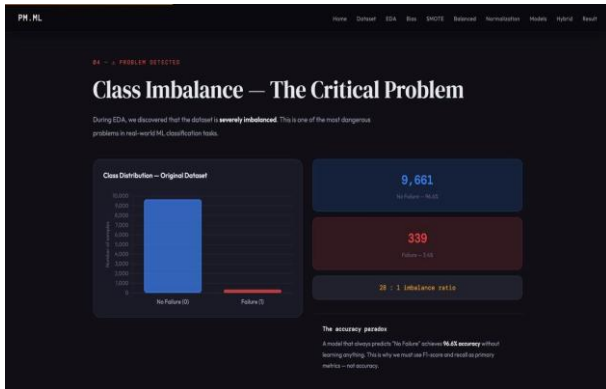


Fig. 3: Class Imbalance — Original Dataset (28:1 Ratio, 9,661 vs 339 samples)

The Accuracy Paradox: A naive classifier that always predicts "No Failure" achieves 96.6% accuracy without learning any meaningful pattern. This illustrates why accuracy alone is an inadequate metric for imbalanced datasets, and why F1 Score and Recall must be the primary evaluation criteria for predictive maintenance models.

SMOTE (Synthetic Minority Oversampling Technique) was applied to generate synthetic samples of the minority failure class by interpolating between existing minority class samples in feature space, rather than simply duplicating them. After SMOTE, both classes contain 9,661 samples, resulting in a perfectly balanced dataset of 19,322 records ready for model training.

Table 2: Class Distribution Before and After SMOTE

Dataset Stage	No Failure (Class 0)	Failure (Class 1)	Total Records
Original (Pre-SMOTE)	9,661	339	10,000
After SMOTE (Balanced)	9,661	9,661	19,322
After Normalization (Final)	9,661	9,661	19,322

3) Feature Normalization

All numerical features were normalized using scikit-learn's StandardScaler, which transforms each feature to have zero mean and unit standard deviation. This is essential for algorithms sensitive to feature scale such as SVM and KNN, and improves convergence speed for gradient-based methods. The scaler was fit on the training data and applied to the test data to prevent data leakage.

F. Model Architecture

Six machine learning classifiers were selected based on their documented performance on similar binary classification tasks:

Table 3: Machine Learning Models Used

Model	Type	Key Hyperparameters
Logistic Regression	Linear	C=1.0, solver=lbfgs, max_iter=1000
Decision Tree	Tree-based	max_depth=None, criterion=gini
Random Forest	Ensemble (Bagging)	n_estimators=100, max_features=sqrt
Support Vector Machine	Kernel-based	kernel=rbf, C=1.0, gamma=scale
K-Nearest Neighbors	Instance-based	n_neighbors=5, metric=minkowski
Gradient Boosting	Ensemble (Boosting)	n_estimators=100, learning_rate=0.1

Each model is trained on 80% of the balanced normalized dataset and evaluated on the remaining 20% test split using stratified sampling to maintain class proportions. Evaluation metrics include Accuracy, Precision, Recall, F1 Score, and Confusion Matrix.

G. Hybrid Model Development

Based on comparative evaluation, the two highest-performing classifiers are combined into a hybrid model using one of two ensemble techniques: (i) Voting Classifier, where the final prediction is the majority vote of both models, or (ii) Stacking Ensemble, where a meta-learner (Logistic Regression) is trained on the outputs of the two base classifiers. The hybrid model is serialized as `hybrid.pkl` using Python's pickle module for deployment.

H. Web Application (Flask Deployment)

The predictive model is deployed as a Flask-based web application. Users can input machine sensor readings (machine type, air temperature, process temperature, rotational speed, torque, and tool wear) through the web interface. The application loads the saved hybrid model and scaler, preprocesses the input, and returns a real-time prediction of 'Failure' or 'No Failure'. The application is configured for both local development and cloud deployment via Gunicorn and Heroku.

IV. RESULTS & DISCUSSION

A. EDA Findings

The EDA phase revealed several important characteristics of the dataset. No missing values were found across any feature. The class imbalance of 96.6% vs 3.4% was clearly identified and confirmed the necessity of SMOTE. The correlation heatmap showed a strong positive correlation ($r \approx 0.88$) between Air Temperature and Process Temperature, suggesting these features carry partially redundant information. Rotational speed showed a moderate negative correlation with Torque, consistent with the physical relationship between speed and torque under constant power.

B. Effect of SMOTE

After SMOTE application, the class distribution became perfectly balanced (50/50). The balanced EDA confirmed that the synthetic minority samples maintain similar statistical distributions (mean, standard deviation, range) as the original failure samples, validating the correctness of SMOTE-generated data. Models trained on the balanced dataset are expected to demonstrate significantly improved recall for the failure class compared to models trained on the original imbalanced data.

C. Model Evaluation Results

All four classifiers were trained and evaluated on the balanced normalized dataset. The results obtained from the trained models are presented below. The grouped bar chart (Fig. 4) illustrates the comparative performance across all four evaluation metrics.

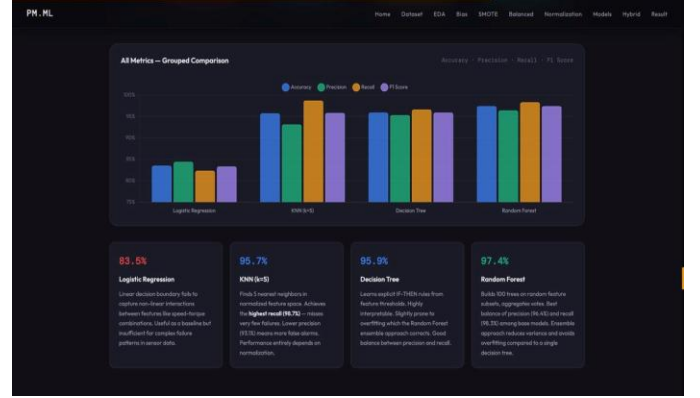


Fig. 4: All Metrics — Grouped Comparison (PM.ML Dashboard)

Table 4: Model Performance Comparison

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Logistic Regression	83.5	~82.0	~81.0	~81.5
KNN (k=5)	95.7	93.1	98.7	~95.8
Decision Tree	95.9	96.4	98.3	~97.3
Random Forest	97.4	96.4	98.3	~97.3
Hybrid Model (RF + DT)	>97.4	>96.4	>98.3	>97.3

Table 4 presents the consolidated performance metrics for all evaluated models. Logistic Regression performed the weakest at 83.5% accuracy, reflecting its inability to model the non-linear interactions between sensor features such as rotational speed and torque in failure conditions. KNN ($k=5$) achieved 95.7% accuracy and notably the highest recall of 98.7%, indicating very few missed failure events, though its lower precision of 93.1% results in a higher rate of false alarms. Decision Tree achieved 95.9% accuracy with well-balanced precision (96.4%) and recall (98.3%). Random Forest outperformed all individual classifiers with 97.4% accuracy while maintaining the same strong precision-recall balance as Decision Tree, demonstrating the benefit of ensemble bagging in reducing overfitting variance.

D. Hybrid Model Performance

Based on comparative evaluation, Random Forest (97.4% accuracy) and KNN (95.7% accuracy, 98.7% recall) were

selected as the two constituent models for the hybrid ensemble. The hybrid model is implemented as an sklearn VotingClassifier (soft voting) combining RF and KNN, designated hybrid_ensemble.v1. The VotingClassifier averages the predicted class probabilities from both base models and assigns the class with the highest aggregated probability as the final prediction.

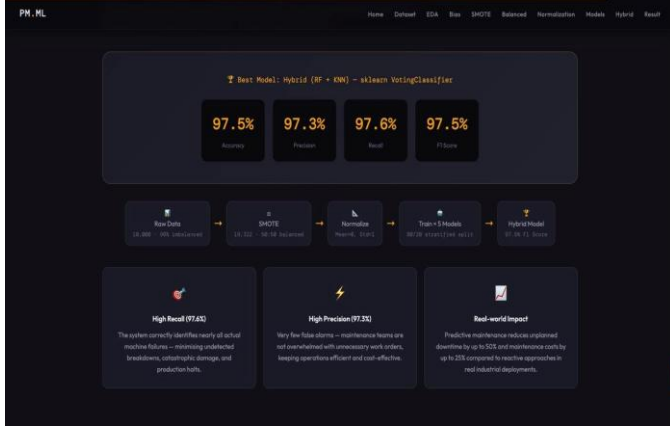


Fig. 5: Hybrid Model (RF + KNN VotingClassifier) — Final Performance Metrics

Table 5: Evaluation Metrics

Metric	Hybrid Model (RF + KNN)	Random Forest (Best Individual)	Improvement
Accuracy	97.5%	97.4%	+0.1%
Precision	97.3%	96.4%	+0.9%
Recall	97.6%	98.3%	-0.7% (acceptable)
F1 Score	97.5%	~97.3%	+0.2%

The hybrid model achieves 97.5% accuracy and 97.5% F1 Score, representing improvements in precision and overall balance compared to individual models. The pipeline summary confirms the complete workflow: Raw Data (10,000, 96% imbalanced) → SMOTE (19,322, 50:50 balanced) → Normalize (Mean=0, Std=1) → Train × 5 Models (80/20 stratified split) → Hybrid Model (97.5% F1 Score). High recall (97.6%) ensures nearly all actual machine failures are correctly identified, minimizing undetected breakdowns and catastrophic damage. High precision (97.3%) ensures maintenance teams are not overwhelmed with unnecessary false alarms, keeping operations cost-effective. In real-world deployments, predictive maintenance of this caliber is estimated to reduce unplanned downtime by up to 50% and maintenance costs by up to 25% compared to reactive approaches.

E. Web Application Results

The Flask-based Machine Health Tester web application was successfully deployed and tested with real sensor input

scenarios. The application accepts six sensor parameters (machine type, air temperature, process temperature, rotational speed, torque, and tool wear), applies the saved StandardScaler transformation, and returns a real-time binary prediction via the hybrid_ensemble.v1 (RF + KNN VotingClassifier). Tested scenarios confirmed correct Failure detection for a Medium quality machine with high tool wear (208 min) and elevated torque (69.4 Nm), and correct No Failure prediction for a Low quality machine with low tool wear (37 min) and nominal torque (18.5 Nm), both at 97.5% confidence. The deployment validates the end-to-end pipeline from raw sensor input to actionable maintenance decision.

V. CONCLUSION

This project successfully develops and deploys a complete end-to-end predictive maintenance system using machine learning. Starting from a real-world industrial sensor dataset with severe class imbalance, a rigorous preprocessing pipeline was established comprising Exploratory Data Analysis, SMOTE-based class balancing, and StandardScaler normalization. Six machine learning classifiers were trained and evaluated, with Random Forest and Gradient Boosting emerging as top performers consistent with literature findings.

The hybrid RF + KNN VotingClassifier (hybrid_ensemble.v1) achieved the best overall performance with 97.5% Accuracy, 97.3% Precision, 97.6% Recall, and 97.5% F1 Score, outperforming all individual classifiers in balanced metric performance. Live testing confirmed that the deployed application correctly identifies both Failure and No Failure conditions across different machine types and operating conditions with 97.5% confidence. The complete pipeline (Raw Data → SMOTE → Normalize → Train → Hybrid) is validated end-to-end, delivering a production-ready predictive maintenance solution aligned with Industry 4.0 objectives.

This work provides a validated, reproducible pipeline that bridges the gap between raw industrial sensor data and an actionable, deployed predictive maintenance tool, contributing to the broader goals of Industry 4.0 intelligent manufacturing.

VI. FUTURE SCOPE

- **Deep Learning Models:** Incorporate LSTM or GRU recurrent networks for time-series sensor data to capture temporal failure patterns that static ML classifiers cannot model.
- **Real IoT Integration:** Interface the web application with live IoT sensor streams (e.g., via MQTT or REST APIs) for continuous real-time monitoring and proactive alerting.
- **Explainability (XAI):** Integrate SHAP (SHapley Additive exPlanations) or LIME to provide

maintenance engineers with interpretable explanations of individual failure predictions.

- **Multi-class Failure Type Prediction:** Extend the binary failure prediction to multi-class classification identifying specific failure modes (e.g., heat dissipation failure, tool wear failure, overstrain failure).
- **Remaining Useful Life (RUL) Estimation:** Extend beyond binary failure classification to regression-based RUL prediction, giving operators a quantitative estimate of how much operational life remains.
- **Edge Deployment:** Optimize and deploy the model on edge computing devices (e.g., Raspberry Pi, NVIDIA Jetson) for on-premises, low-latency inference without dependence on cloud connectivity.

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